HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 11 Name: _____ Marks: / 36

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

   A. $x > 9$

   B. $x > 10$

   C. $x > 11$

   D. $x > 12$

(DSE-2013 PII Q05)

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2. Let k be a constant. Solve $(x-k)^2 = 4k^2$.

A. $x = 3k$

B. $x = 5k$

C. $x = -k$ or $x = 3k$

D. $x = -3k$ or $x = 5k$

(DSE-2013 PII Q06)

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3. A sum of $50 000 is deposited at 8% per annum for 1 year, compounded monthly. Find the interest to the nearest dollar.

A. $4 000

B. $4 122

C. $4 143

D. $4 150

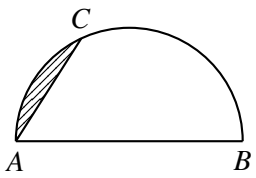
(DSE-2013 PII Q11)

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4. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,cm^2$.



A. $0.23\,cm^2$

B. $0.52\,cm^2$

C. $0.64\,cm^2$

D. $1.07\,cm^2$

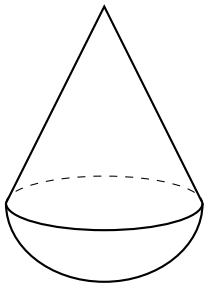
(DSE-2013 PII Q16)

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5. In the figure, the solid consists of a right circular cone and a hemisphere with a common base. The base radius and the height of the circular cone are 3 cm and 4 cm respectively. Find the total surface area of the solid.



A. $30\pi\,\text{cm}^2$

B. $33\pi\,\text{cm}^2$

C. $48\pi\,\text{cm}^2$

D. $51\pi\,\text{cm}^2$

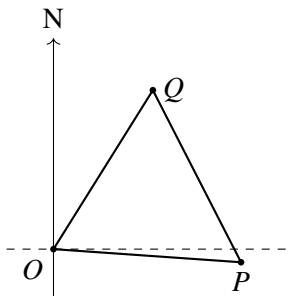
(DSE-2013 PII Q17)

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6. In the figure, the bearing of P from O is $S86^\circ E$ and the bearing of from O is $N32^\circ E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

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7. In $\triangle ABC$, $AB : BC : AC = 8 : 15 : 17$. Find $\cos A : \cos C$.

A. 8 : 15

B. 8 : 17

C. 15 : 8

D. 15 : 17

(DSE-2013 PII Q22)

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8. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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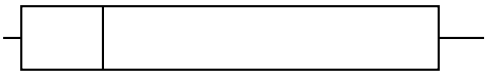
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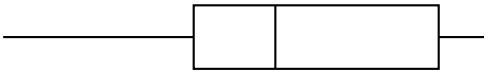
9. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

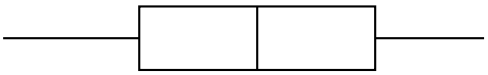
A.



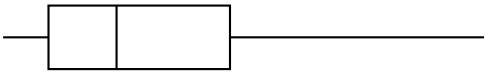
B.



C.



D.



(DSE-2013 PII Q29)

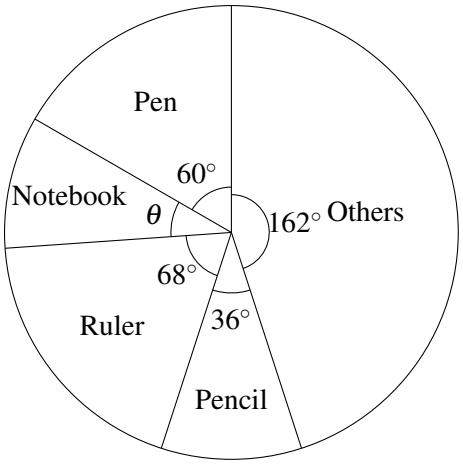
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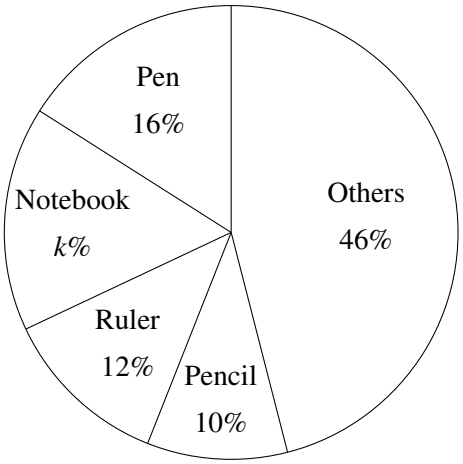
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10. The pie charts below show the distributions of the profits of stationery shop$X$ and stationery shop Y from the sales of stationery in a certain month. Which of the following must be true?

Distribution of the profits of stationery shop X



Distribution of the profits of stationery shop Y



A. In that month, the profit from the sales of pencils of stationery shop X is the same as that of stationery shop Y.

B. In that month, the total profit from the sales of pens and notebooks of stationery shop X is less than the total profit from the sales of rulers and pencils of the shop.

C. $k = 14$

D. $\theta = 36^\circ$

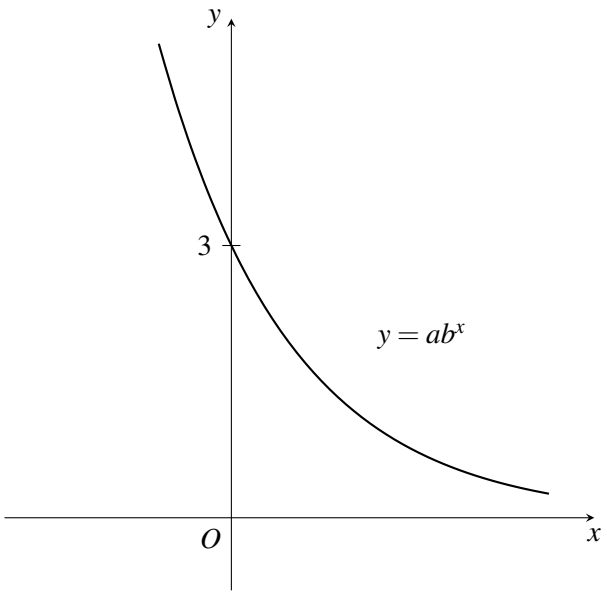
(DSE-2013 PII Q30)

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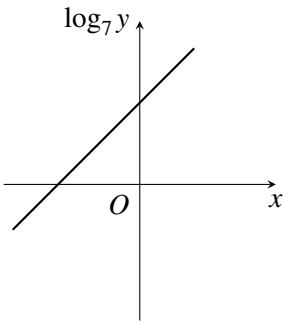
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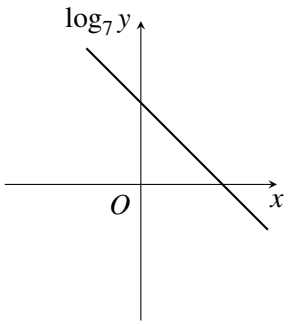
11. The figure below shows the graph of $y = ab^x$, where a and b are constants. Which of the following graphs may represent the relation between x and $\log_7 y$?



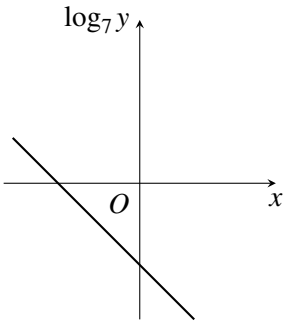
A.



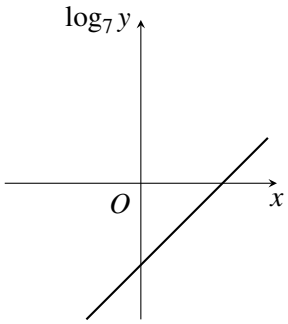
B.



C.



D.



(DSE-2013 PII Q32)

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12. $\text{A00000E00001}_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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13. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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14. If $\alpha \neq \beta$ and $\begin{cases} 3\alpha = \alpha^2 - 5 \\ 3\beta = \beta^2 - 5 \end{cases}$, then $\alpha\beta =$

A. 3

B. -3

C. 5

D. -5

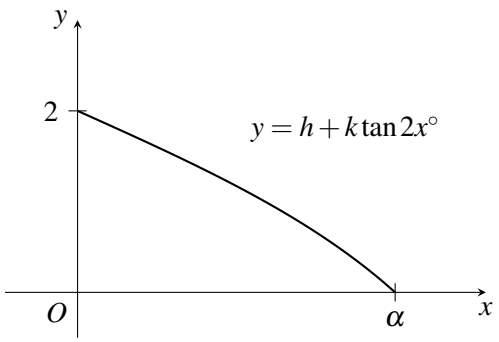
(DSE-2013 PII Q35)

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15. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

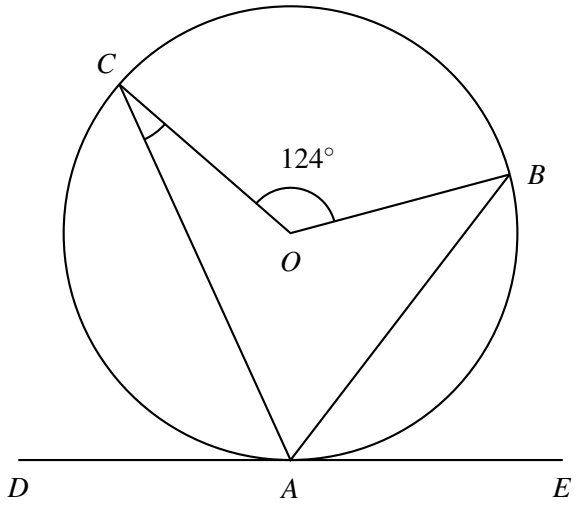
(DSE-2013 PII Q39)

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16. In the figure, O is the centre of the circle ABC, and DE is tangent to the circle at A. If AB is the angle bisector of $\angle CAE$, then $\angle ACO =$



A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

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17. Find the range of k such that the circle $x^2 + y^2 + 2x - 2y - 7 = 0$ and the straight line $3x - 4y + k = 0$ intersect.

A. $-8 < k < 22$

B. $-8 \leq k \leq 22$

C. $k < -22$ or $k > 8$

D. $k \leq -22$ or $k \geq 8$

(DSE-2013 PII Q42)

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18. Let O be the origin. If the coordinates of the points A and B are $(0, 12)$ and$(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of ΔOAB is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

I can finish _____ questions

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